We introduce a pair of smooth functions $a(x,t)$ and $b(x,t)$, where a represents a drift tendency and b the average magnitude of the random jump. In the interval $[n\delta t,(n+1)\delta t]$, a step $\delta \mathrm{X}_n = \mathrm{X}_{n+1} - \mathrm{X}_n$ of the random walk will be given by

$$\delta \mathrm{X}_n = a(\mathrm{X}_n, t_n)\, h + b(\mathrm{X}_n, t_n)\mathrm{J}_n\, \delta x, \tag{6.9}$$

where the random increment J_n was introduced in Sect. 6.2.1. Choose a positive t and sum (6.9) from $n=0$ to $\nu-1$, where $\nu = \lfloor t/\delta t \rfloor$, obtaining

$$\mathrm{X}_\nu - \mathrm{X}_0 = \sum_{n=0}^{\nu-1} \delta \mathrm{X}_n = \sum_{n=0}^{\nu-1} a\,(\mathrm{X}_n, t_n)\,\delta t + \sum_{n=0}^{\nu-1} b(\mathrm{X}_n, t_n)\mathrm{J}_n \delta x. \tag{6.10}$$

Take the diffusion limit $\delta t \to 0$, $\delta x^2/\delta t \to 1$ with $\nu \delta t = t$ fixed, then we obtain, under the Itō convention,

$$\mathrm{X}(t) - \mathrm{X}(0) = \int_0^t a(\mathrm{X}(s), s)\mathrm{d}s + \int_0^t b(\mathrm{X}(s), s)\,\mathrm{dW}(s), \tag{6.11}$$

where X is obtained by a limiting process similar to that discussed above, and

$$\int_0^t a(\mathrm{X}(s), s)\mathrm{d}s \stackrel{\text{def}}{=} \lim_{\delta t \to 0} \sum_{n=0}^{\nu-1} a\,(\mathrm{X}(n\delta t), n\delta t)\,\delta t, \tag{6.12}$$

$$\int_0^t b(\mathrm{X}(s))\mathrm{dW}(s) \stackrel{\text{def}}{=} \text{l.i.m.}_{\delta t \to 0} \sum_{n=0}^{\nu-1} b\,(\mathrm{X}(n\delta t), n\delta t)\,\mathrm{J}_k \sqrt{\delta t}. \tag{6.13}$$

As $\mathrm{X}(t) - \mathrm{X}(0) = \int_0^t \mathrm{dX}(s)$, we may view (6.11) as a relation between differentials:

$$\mathrm{dX}(s) = a(\mathrm{X}(s), s)\mathrm{d}s + b(\mathrm{X}(s), s)\,\mathrm{dW}(s),$$

or, more simply

$$\mathrm{dX} = a(\mathrm{X}, t)\mathrm{d}t + b(\mathrm{X}, t)\,\mathrm{dW}. \tag{6.14}$$

which is the general form of a scalar stochastic differential equation.

We mention in passing the extension to systems of SDEs. Let $\boldsymbol{a}(\boldsymbol{z},t)$ and $\boldsymbol{b}^{(i)}(\boldsymbol{z},t)$, $i = 1,2,\ldots,r$ be a set of $r+1$ vector fields in $\mathbb{R}^m$, and let $\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_r$ be a

collection of r independent Wiener processes, then we may write a system of SDEs in $\mathbb{R}^m$ in the form

$$\mathrm{d}\mathbf{Z} = \boldsymbol{a}(\mathbf{Z}, t)\mathrm{d}t + \sum_{i=1}^{r} \boldsymbol{b}^{(i)}(\mathbf{Z}, t)\,\mathrm{dW}_i. \tag{6.15}$$

alternatively, if $\boldsymbol{B}$ is the matrix with columns $\boldsymbol{b}^{(1)}, \boldsymbol{b}^{(2)}, \ldots, \boldsymbol{b}^{(r)}$, and $\mathbf{W}$ represents the r-component vector of Wiener processes, then

$$\mathrm{d}\mathbf{Z} = \boldsymbol{a}(\mathbf{Z}, t)\mathrm{d}t + \boldsymbol{B}(\mathbf{Z}, t)\,\mathrm{d}\mathbf{W}. \tag{6.16}$$

6.3.1 The Itō Formula

We shall derive, heuristically, a core piece of mathematical apparatus which permits much of the analysis of SDEs. Let $\phi(\cdot)$ be any continuously twice differentiable function and let $\mathrm{X}_n \in \mathbb{R}$ represent the state of a biased random walk as considered in Sect. 6.3, i.e., with state $\mathrm{X}_{n+1} = \mathrm{X}_n + a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x$. Let $\phi_n = \phi(\mathrm{X}_n)$, then Taylor's theorem gives

$$\begin{aligned}
\delta\phi &\stackrel{\text{def}}{=} \phi_{n+1} - \phi_n \\
&= \phi'(\mathrm{X}_n)\delta\mathrm{X}_n + \frac{1}{2}\phi''(\mathrm{X}_n)\delta\mathrm{X}_n^2 + o(\delta\mathrm{X}_n^2) \\
&= \phi'(\mathrm{X}_n)(a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x) \\
&\qquad + \frac{1}{2}\phi''(\mathrm{X}_n)(a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x)^2 + o(\delta\mathrm{X}_n^2).
\end{aligned}$$

Now we expand the 2nd order term and, working in the diffusion limit, recognize that $\delta x^2 \sim \delta t$, thus we have

$$\begin{aligned}
\delta\phi &= \phi'(\mathrm{X}_n)(a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x) \\
&\qquad + \frac{1}{2}\phi''(\mathrm{X}_n)b(\mathrm{X}_n, n\delta t)^2\delta t + \mathcal{O}(\delta t^{3/2}).
\end{aligned}$$

Passing to the limit, this leads to the SDE:

$$\mathrm{d}\phi(\mathrm{X}) = \phi'(\mathrm{X})(a(\mathrm{X}, t)\mathrm{d}t + b(\mathrm{X}, t)\mathrm{dW}) + \tfrac{1}{2}\phi''(\mathrm{X})b(\mathrm{X}, t)^2\mathrm{d}t. \tag{6.17}$$